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Roll no. : 25p0014

Class : BAI-2A

Labtask : 10

Question 1 :

Code :

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4  class Distance
5  {
6  private:
7      int inch;
8      int feet;
9
10 public:
11
12     Distance() : feet(0), inch(0) {};
13     Distance(int f, int i) : feet(f), inch(i) {};
14
15     Distance operator + (Distance &D)
16     {
17         Distance resultant;
18         resultant.inch = inch + D.inch;
19         resultant.feet = feet + D.feet;
20         return resultant;
21     }
22
23     void feet_to_inch(int &f, int &i)
24     {
25         f += inch/12;
26         inch = inch%12;
27     }
28
29
30     void setdata(int f, int i)
31     {
32         feet= f;
33         inch = i;
34     }
```

```

35
36 void showData()
37 {
38     feet_to_inch(feet, inch);
39     cout<<"Feet : "<<feet<<endl;
40     cout<<"Inches : "<<inch<<endl;
41 }
42
43 };
44
45 int main()
46 {
47     Distance D1;
48     Distance D2(2, 14);
49     D1.setdata(1,24);
50
51     Distance D3 = D1 + D2;
52     D3.showData();
53
54     return 0;
55 }

```

Output:

```

syeda areej-e-fa
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Feet : 6
Inches : 2
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```

Question 2 :

Code :

```
lab_tasks > labtask10 > Q2.cpp > main()
1 > #include<iostream>...
3 using namespace std;
4 class student
5 {
6     public:
7     char* name;
8     int roll_no;
9
10    student(const char* n, int r)
11    {
12        name = new char[strlen(n)+1];
13        strcpy(name, n);
14        roll_no = r;
15    }
16
17
18    void showData()
19    {
20        cout<<"Name : "<<name<<endl;
21        cout<<"Roll no: "<<roll_no<<endl;
22    }
23
24
25    ~student()
26    {
27        delete[] name;
28        cout<<"destructor called"<<endl;
29    }
30
31 };
32
33 int main()
34 {
35     student s1("Usman", 101);
36     student s2 = s1;
37
38     s1.showData();
39     s2.showData();
40
41     return 0;
42 }
```

Output:

```
_lab/lab_tasks/labta
Name : Usman
Roll no: 101
Name : Usman
Roll no: 101
destructor called
```

Question 3 :

Code :

```
lab_tasks > labtask10 > Q3.cpp > student
1  #include<iostream>
2  #include<cstring>
3  using namespace std;
4  class student
5  {
6      public:
7      char* name ;
8      int roll_no;
9
10     // parameterized copy constructor
11     student(const char* n, int r)
12     {
13         name = new char[strlen(n)+1];
14         strcpy(name, n);
15         roll_no = r;
16     }
17
18     // deep copy constructor
19     student(const student&S)
20     {
21         name = new char[strlen(S.name)+1];
22         strcpy(name, S.name);
23         roll_no = S.roll_no;
24     }
25
26     // display function
27     void showData()
28     {
29         cout<<"Name : "<<name<<endl;
30         cout<<"Roll no: "<<roll_no<<endl;
31     }
32
```

```

33     void update_name(char *n)
34     {
35         delete[] name;
36         name = new char[strlen(n)+1];
37         strcpy(name, n);
38     }
39
40
41     ~student()
42     {
43         delete[] name;
44         cout<<"destructor called"<<endl;
45     }
46
47
48 };
49 int main()
50 {
51     student s1("Usman", 101);
52     student s2 = s1;
53
54     cout<<"----Before update----"<<endl;
55     s1.showData();
56     s2.showData();
57
58     s2.update_name("Ali");
59
60     cout<<"----After update----"<<endl;
61     s1.showData();
62     s2.showData();
63
64
65     return 0;
66 }

```

Output:

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----Before update----

Name : Usman

Roll no: 101

Name : Usman

Roll no: 101

----After update----

Name : Usman

Roll no: 101

Name : Ali

Roll no: 101

destructor called

destructor called

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